Can the review teach a move from his lessons instead of the engine's top line?

The lesson pick at the player's turning points

Chess Coach research lab

Abstract

At each of the player's turning points the daily review teaches a better move and the line from it. He asked for the move his study lessons teach, whenever the engine says it holds, named with the lesson's rule, instead of the engine's top line. The app's lesson pick chooses among the review's stored engine lines that hold for him: within 3 win% of the engine's first line, or keeping a won game won. Among those it takes the move a lesson teaches for a purpose his own move lacked, reading each move's purpose with the board code. At 1,023 turning points up to 2026-09-16 and 488 after, a lesson taught the pick at 77.3% and 71.1%. The pick gave up 0.32 (0.26 to 0.38) and 0.37 (0.29 to 0.46) win% against the engine's first line, and never left the holding verdicts. Players at his band choose it more often than the engine's first line, by 3.0 (2.2 to 3.9) and 3.6 (2.5 to 4.8) points of Maia-2's probability. By the rule fixed before the run, the pick is kept. It is the engine's first line at about 74% of turning points; where it is not, his band chooses it 11.9 (8.7 to 15.2) and 14.0 (10.0 to 18.4) points more often. The lessons it names are few: 21 of the 74, and two chapters carry 55% of the lesson picks, which shows where the next round should go.

1 The question

The daily review teaches, at each turning point, a better move and the line from it: the engine's first line, or the move players at his band would likely play among those the engine says hold. Since the app has study lessons (74 of them, each a video and a written lesson whose recommended moves were checked with Stockfish), the player asked for that move to come from what the lessons teach whenever the engine says it holds: a queen trade, a rook to an open file, a pawn break, a defence that meets the threat, named with the lesson's rule.

Two earlier studies bound what such a pick may claim. A rule of thumb explains a move but should not choose it (rules-of-thumb), and a sound move he would find costs about what the engine says it gives up (findable-move). So the lessons only choose among moves the engine already says hold. At his turning points, how often does such a move exist, what does it cost against the engine's first line, and do players at his band choose it more often than that first line?

2 How it was measured

The positions. Every turning point in the app's archive with him to move and at least one stored engine line: 1,511 turning points. The seven games of the two days read while the change was

designed are left out. The candidates are the first moves of the stored lines, up to five, from the routine's deep search, each with its stored evaluation in his view. No new engine search was run.

What holds. Each candidate is graded against the engine's first line, E, by the review's own grading. It *holds* for him when it is *best*, within 3 win% of E (the review's window), or *keeps* the result: E's position is won (at least 88 win%, or a mate) and the candidate's still is.

His band. The Maia-2 level of his rating at the game with the app's offset (+250), and a move's *share* is Maia-2's probability for it at that band. N, the natural move, is the holding candidate with the largest share.

The pick. The board code reads what each move is for, from the move and its line: it meets a threat, wins material, trades queens, takes an open file with a rook, breaks with a pawn, improves the worst piece, and so on. The study service reads the same purposes from every move the lessons recommend, and for a purpose offers the lesson chapters whose lesson is tagged with one of its ideas, ranked by how well the chapter's own recommended moves match it. L, the lesson pick, is the holding candidate his band chooses most often among those a lesson teaches for a purpose his move lacked (`lesson`); else among those a lesson teaches (`lesson-shared`); else the natural move; else E. His move's purposes are read with the line the routine stored to punish it. The code is the app's at commit `f7a8b12a01`.

The rule, fixed before the run. Train: the turning points up to 2026-09-16, the first two thirds. Held out: the rest, up to 2026-11-02. On both, three bars. *Coverage*: a lesson teaches L (`lesson` or `lesson-shared`) at 50% of turning points or more. *Cost*: L's mean win% loss against E is at most 1.5, and no L falls outside the holding verdicts. *Findability*: L's share minus E's, paired by turning point, has its 95% interval above zero, from 10,000 bootstrap draws over games (a game's turning points move together). The pick is kept when all three pass on both.

3 The verdict

	Train	Held out
Turning points (games)	1,023 (399)	488 (191)
Taught by a lesson (bar: 50%)	**77.3%**	**71.1%**
for a purpose his move lacked	63.7%	58.0%
Loss against E, win% (bar: 1.5)	**0.32 (0.26 to 0.38)**	**0.37 (0.29 to 0.46)**
largest; picks outside the holding verdicts	10.2; 0	11.1; 0
Share of L minus share of E, points (bar: above 0)	**3.0 (2.2 to 3.9)**	**3.6 (2.5 to 4.8)**

Table 1: The rule's three measures on each split. Intervals are 95% percentiles of 10,000 bootstrap draws over games. A share is Maia-2's probability for the move at his band, so a point is one percent of his band's choices.

All three bars pass on both splits (Table 1), so the pick is kept. A lesson teaches the pick at most turning points, and at most of them for a purpose his own move lacked. The pick gives up 0.32 and 0.37 win% on average. The largest losses, 10.2 and 11.1 win%, come from the 41 picks that keep a

won game won; every pick that loses more than the window is one of them (nine in all). Players at his band choose the pick a few points more often than the engine's first line, on both splits.

4 What the pick looks like

The pick	Train	Held out
meets a threat	285	122
a rook to an open file	111	70
wins material	127	46
develops	59	16
mates	47	10
a pawn break	29	19
trades queens	26	18
improves the worst piece	23	8
offers a queen trade	17	10
pushes a passed pawn	15	7
castles	17	5
takes the centre	12	4
an ending technique	7	4
a rook to the seventh rank	5	5
takes an outpost	6	1
trades when ahead	5	1
trades into a pawn ending	0	1
Taught by a lesson, all purposes	791	347

Table 2: The purpose a lesson teaches the pick for, as the board code reads it from the move and its engine line. These counts describe; they carry no rule.

These numbers describe; none of them carries a rule.

Mostly the engine's move. L is E at 74% of train turning points and 74% of held-out ones: a lesson teaches the engine's own first line at 68% and 59%. Where L is not E (261 and 127 turning points), his band chooses L 11.9 (8.7 to 15.2) and 14.0 (10.0 to 18.4) points more often than E. The average gain in Table 1 is this gain spread over every turning point.

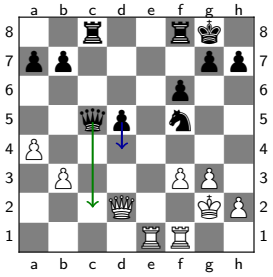
A little less natural than the natural move. L is N at 87% of turning points on both splits, and its share is -2.8 (-3.5 to -2.2) and -2.3 (-3.0 to -1.6) points below N's: choosing the move a lesson teaches, rather than the one his band likes best, costs a little findability.

Defence first. The pick is most often taught for meeting a threat, 36% of the lesson picks, then for a rook on an open file and for winning material (Table 2).

Few chapters. The 1,138 lesson picks name 21 lessons and 26 chapters. One chapter carries 39% of them: "Move the queen while taking the loose pawn" from "Choose a defense that changes the threat", named for meeting a threat and for winning material alike, though 77% of its picks do not move the queen. With "Move the rook that has no job" from "Give your rook an entry square" the

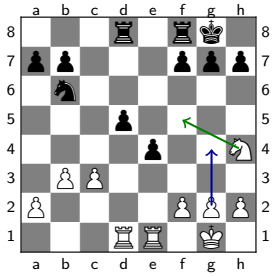
two carry 55%. The ranking reads only the ideas a purpose shares with a chapter's tags, so one chapter stands for most defences.

5 Four positions



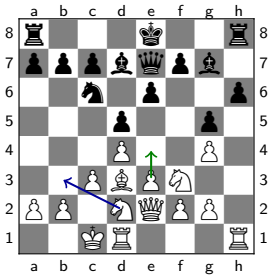
FEN `2r2rk1/pp4pp/5p2/2qp1n2/P7/1P3PP1/3Q2KP/4RR2 b - - 4 31`

(a) Black to move. The pick (green) is 31... ♕c2 (`c5c2`), chosen by 40% of his band; the engine's first line (blue) is 31... d4 (`d5d4`), at 24%; the pick gives up 0.8 win%. The board code: "its line trades queens, and with fewer pieces the extra material counts for more". The chapter: "Clear the passer's path and offer the queens" from "Trade for a reason and inspect what remains". He played 31... ♕a3 (`c5a3`).



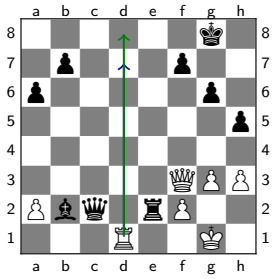
FEN `3r1rk1/pp3ppp/1n6/3p4/4p2N/1PP5/P4PPP/3RR1K1 w - - 1 22`

(b) White to move. The pick is 22. ♘f5 (`h4f5`), at 66%; the engine's first line is 22. g4 (`g2g4`), at under 1%; the pick gives up 1.1 win%. The board code: "it brings the knight on h4, the piece with the fewest moves (3), to f5, where it has 8". The chapter: "A central knight needs something to do" from "Improve the piece with the weakest job". He played 22. c4 (`c3c4`).



FEN `r3k2r/pppbqpb1/2n1p2p/3p2p1/3P2P1/2PBPN2/PP1NQPP1/2KR3R w kq - 4 13`

(c) White to move. The pick is 13. e4 (`e3e4`), at 35%; the engine's first line is 13. ♘b3 (`d2b3`), at 7%; the pick gives up 0.5 win%. The board code: "a pawn break: the e-pawn strikes at d5, and trades there open lines". The chapter: "Attack the supporting pawn" from "Read the pawn structure before choosing a plan". He played 13. ♘e5 (`f3e5`).



FEN `6k1/1p3p2/p5p1/7p/8/5QPP/Pbq1rP2/3R2K1 w - - 3 30`

(d) White to move. The pick is 30. ♖d8+ (`d1d8`), at 82%; the engine's first line is 30. ♖d7 (`d1d7`), at 4%; the pick gives up 0.1 win%. The board code: "it moves the rook on d1 out of attack, to d8". The chapter named, "Move the queen while taking the loose pawn", moves the queen and takes a pawn; the pick does neither. He played 30. ♔f1 (`g1f1`).

Figure 1: The pick in green, the engine's first line in blue, each with the share of his band that chooses it and the chapter the study service ranks first for the pick's purpose. In the last the pick is a defence and the lesson is on defence, but the chapter does not fit the move.

The four positions illustrate the summary; they test nothing. Each is the held-out turning point where the pick, taught for a purpose his move lacked and not the engine's first line, gains most

share over that first line: among the queen trades, the worst piece improved or an outpost taken, the pawn breaks, and the picks naming the most named chapter that do not move the queen.

6 What the app can take

- The pick goes into the review's `teach`: among the moves that hold for him, the move a lesson teaches, with what it does on the board and the chapters that teach it. The daily video can then teach that move and its rule instead of the engine's verdict. Whether the video coaches better for it is a separate study (lesson-videos).
- The chapter a pick names needs a fit check. Ranking chapters by the ideas their tags share with a purpose lets one chapter stand for most defences; a next round could rank them by how closely the chapter's own positions match this one: the piece that moves, what it captures, the threat it meets.
- At 22.7% and 28.9% of turning points no holding move is one a lesson teaches. New purposes for the board code (an attack on the king, prophylaxis, a weak square) would let the lessons teach more of them.

7 Caveats

- **Only the stored lines.** The candidates are the up to five moves the review kept. The live `teach` also weighs his band's likeliest moves.
- **The board code's purposes.** A move is taught for what the board code reads in it; a move can do more than its purposes say, or less.
- **A lesson teaches a move** when a chapter's tags share an idea with the move's purpose. That says the lesson is on the right topic, not that the chapter's rule explains this position, as the last example shows.
- **Share is not understanding.** Maia-2's share says how often players at his band choose a move, not whether he would follow the lesson behind it.
- **The cost is the engine's.** In human hands a sound move he would find costs about what the engine says (findable-move); this study did not play the picks on.